Comparative Scientific Report on Respiratory Particle Deposition Trends

Geometries SD_02, SD_03, and SD_04

Project-style analysis prepared from the uploaded deposition workbook and trend plots

Executive summary

- Across all three geometries, increasing particle diameter shifted transport away from the lower-airway fraction and toward upper-airway capture. The upper-airway crossover occurred near 11.5–13.0 μm at 5 Pa and shifted down to 7.5–8.0 μm at 20 Pa.
- SD_04 showed the strongest nasopharyngeal filtering behavior. Its mean nasopharyngeal fraction was 4.90% at 5 Pa and remained higher than the other two geometries at every pressure level examined.
- SD_02 showed the weakest nasopharyngeal participation and the most direct redistribution from downstream penetration to nasal deposition. SD_03 behaved as an intermediate geometry, but displayed the sharpest pressure sensitivity near 15 Pa.
- Because the workbook label 'lower airway deposition' behaves like a downstream penetration fraction, the report interprets that column cautiously as a lower-airway/penetrating fraction rather than confirmed wall deposition.

1. Objective and data basis

This report compares how aerosol particles distribute within the respiratory tract for geometries SD_02, SD_03, and SD_04 over particle diameters from 0.5 to 20 μm and pressure drops of 5, 10, 15, and 20 Pa. The analysis was performed from the uploaded Excel workbook, which provides geometry descriptors, flow rates, and regional fractions for the nasal cavity, nasopharynx, and lower-airway region for each diameter-pressure condition.

The uploaded PDF plots of nasopharyngeal deposition and escaped particles were used as qualitative cross-checks of the workbook trends. Since the total number of tracked particles differs across geometries, normalized regional fractions from the workbook are treated as the primary basis for between-geometry comparison, while raw count plots are used mainly to confirm the direction and shape of the observed trends.

2. Analysis method

For each geometry and pressure level, the regional fractions were examined as functions of particle diameter. Three summary metrics were then extracted: (i) the mean regional fraction across the full diameter range, (ii) the transition diameter at which upper-airway capture (nasal cavity + nasopharynx) first exceeded the lower-airway fraction, and (iii) the particle size at which nasopharyngeal deposition reached its maximum. These metrics provide a compact way to compare how anatomy modifies the onset, location, and intensity of deposition.

A practical note is necessary. The workbook column labeled 'lower airway deposition' behaves as a near-complementary fraction to upper-airway capture and therefore resembles downstream penetration more than direct lower-airway wall deposition. The results are still physically useful for comparison, but the nomenclature should be clarified in future exports so that deposition and penetration are not conflated.

3. Geometry and flow context

The three geometries are not dynamically equivalent at the same pressure drop. Flow rate rises with pressure in all cases, but SD_02 consistently supports the highest volumetric flow, whereas SD_04 is generally the most restrictive. SD_04 also has the largest nasopharyngeal surface area and volume in the workbook, which is consistent with its stronger mid-size nasopharyngeal capture.

Table 1. Geometry descriptors and flow-rate response extracted from the workbook.

Geometry	Tracked particles	Penetrative depth	Nasopharynx area	Nasopharynx volume	Total cavity volume	Q at 5 Pa	Q at 10 Pa	Q at 15 Pa	Q at 20 Pa
SD_02	1280	96.0	2438.7	4815.4	100138	14.00	20.33	25.45	29.80
SD_03	1644	108.6	3924.0	6347.9	131389	12.49	18.41	20.70	27.01
SD_04	2494	105.3	5067.9	7941.8	120599	10.85	16.50	20.97	24.83

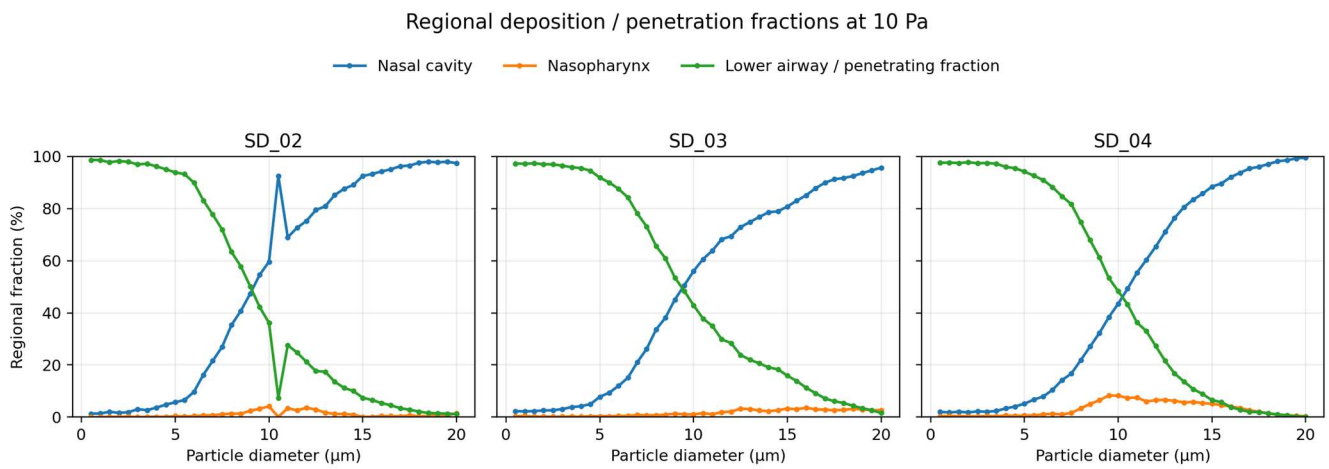

Figure 1. Regional fraction versus particle diameter at 10 Pa. All three geometries show a shift from lower-airway penetration at small diameters to dominant nasal capture at larger diameters, but the transition pathway differs by geometry.

4. Results and discussion

4.1 Common behavior across all three geometries

A common size-dependent pattern appears in every geometry. At the smallest diameters, particles predominantly remain in the lower-airway fraction, indicating strong penetration through the upper airway. As diameter increases, inertial effects become more important and the regional balance shifts upward. By the large-diameter end of the tested range, nasal deposition dominates in all three cases. This is the central trend of the dataset and it persists regardless of geometry.

Pressure intensifies this shift. The diameter at which upper-airway capture first exceeds the lower-airway fraction moves from approximately 11.5–13.0 μm at 5 Pa to 7.5–8.0 μm at 20 Pa. In other words, stronger inhalation conditions pull the deposition transition toward smaller particles.

4.2 Mean regional redistribution with pressure

The pressure-averaged behavior is especially clear when the full diameter range is summarized by mean regional fractions. For all three geometries, the mean nasal fraction increases with pressure, while the mean lower-airway fraction decreases. In contrast, the mean nasopharyngeal fraction tends to drop with pressure after the 5 Pa case, implying that stronger flow does not simply amplify mid-airway filtering; rather, it promotes a more direct transfer from downstream penetration to upstream nasal capture.

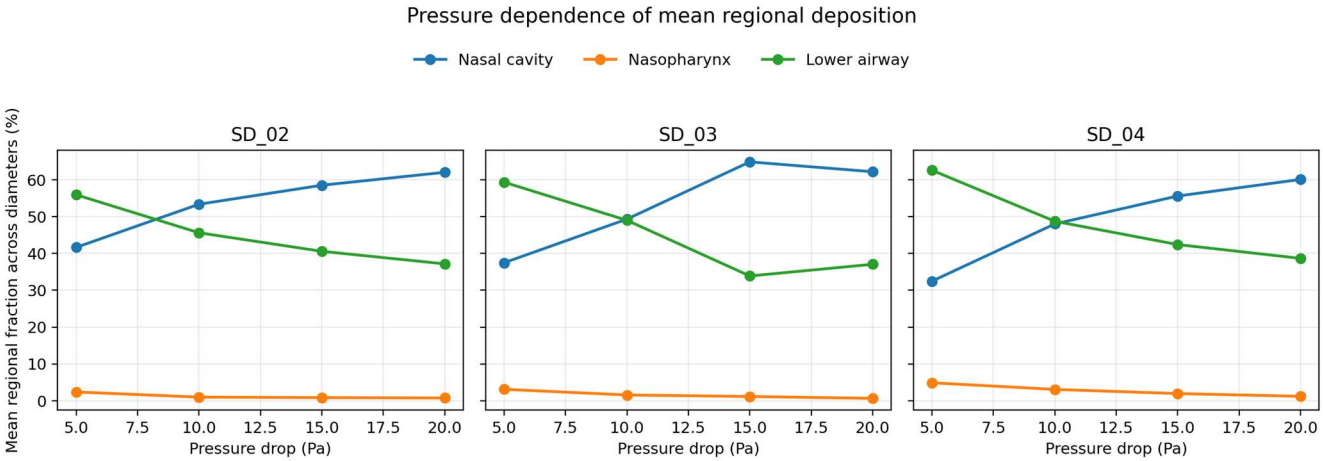

Figure 2. Mean regional fractions across the full diameter range. Increasing pressure consistently raises nasal deposition and reduces the lower-airway fraction in all three geometries.

4.3 Geometry-specific interpretation

SD_02 shows the weakest nasopharyngeal role overall. Its mean nasopharyngeal fraction falls from 2.40% at 5 Pa to 0.77% at 20 Pa, and its peak nasopharyngeal fraction never exceeds about 6.56%. The geometry therefore behaves like a system in which particles either continue downstream when they are small or are captured nasally once they become sufficiently inertial, with relatively little residence in the middle section.

SD_03 is the most balanced of the three geometries. At 10 Pa its mean nasal and lower-airway fractions are nearly equal (49.32% and 49.00%, respectively), and at 15 Pa it exhibits the smallest transition diameter of the set, only 6.5 μm. This suggests stronger sensitivity to the pressure increase around the mid-range conditions.

SD_04 exhibits the strongest nasopharyngeal filtering signature. It has the highest mean nasopharyngeal fraction at every pressure level, reaching 4.90% at 5 Pa, and its peak nasopharyngeal fraction rises to 11.55% at 16 μm. Even at 10 Pa, SD_04 still retains an 8.30% nasopharyngeal peak at 9.5 μm. This geometry therefore appears to create the most effective mid-airway interception of intermediate particles.

Table 2. Key comparative metrics used to summarize each pressure condition.

Geometry	Pressure (Pa)	Upper > lower transition (μm)	Nasopharyngeal peak diameter (μm)	Nasopharyngeal peak fraction (%)
SD_02	5	11.5	14.0	6.56
SD_02	10	9.5	10.0	4.14
SD_02	15	8.0	10.5	3.98
SD_02	20	7.5	7.5	2.34

SD_03	5	12.0	16.0	7.18
SD_03	10	9.5	16.0	3.59
SD_03	15	6.5	12.5	2.92
SD_03	20	7.5	14.5	2.43
SD_04	5	13.0	16.0	11.55
SD_04	10	10.0	9.5	8.30
SD_04	15	9.0	8.5	5.21
SD_04	20	8.0	11.0	3.89

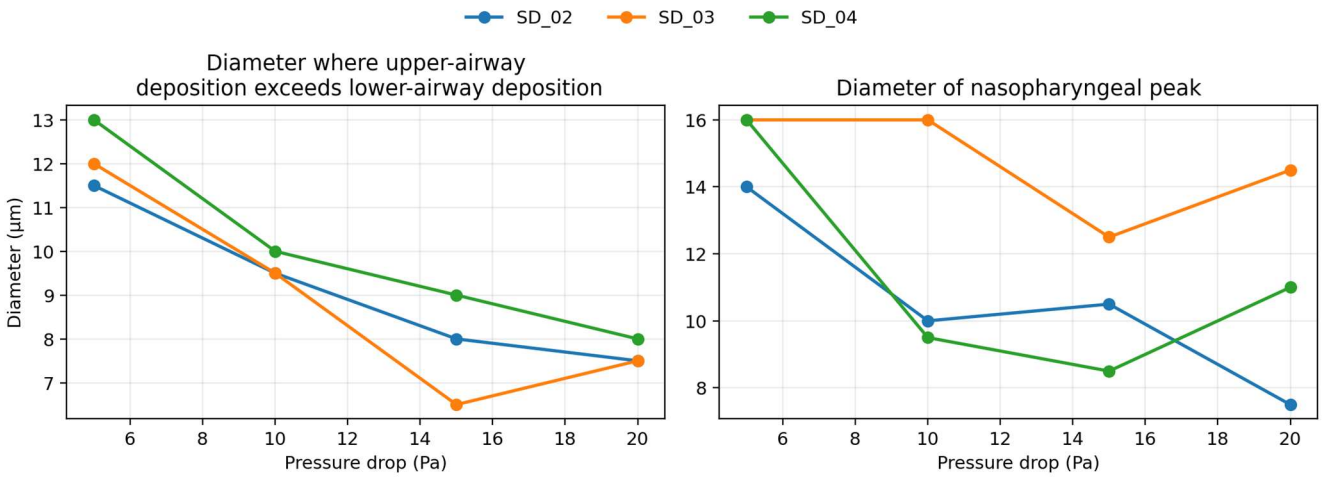

Figure 3. Transition size and nasopharyngeal peak size. Increasing pressure shifts the global upper-airway transition to smaller diameters, but the geometry dependence of the nasopharyngeal peak remains non-monotonic.

4.4 Cross-check against uploaded count plots

The uploaded nasopharyngeal count plots support the same qualitative ranking seen in the normalized workbook data. SD_04 has the largest absolute nasopharyngeal counts, SD_03 is intermediate, and SD_02 is lowest. The uploaded escaped-particle plots also show a consistent monotonic decrease with particle diameter, with the curves shifting left as pressure increases. That behavior agrees with the fraction-based conclusion that increasing pressure suppresses downstream penetration for smaller and smaller particles.

Direct magnitude comparison of raw counts should still be made carefully because the number of injected particles differs across geometries. For this reason, the present report uses the count plots mainly as trend validation, not as the main normalization basis for geometry ranking.

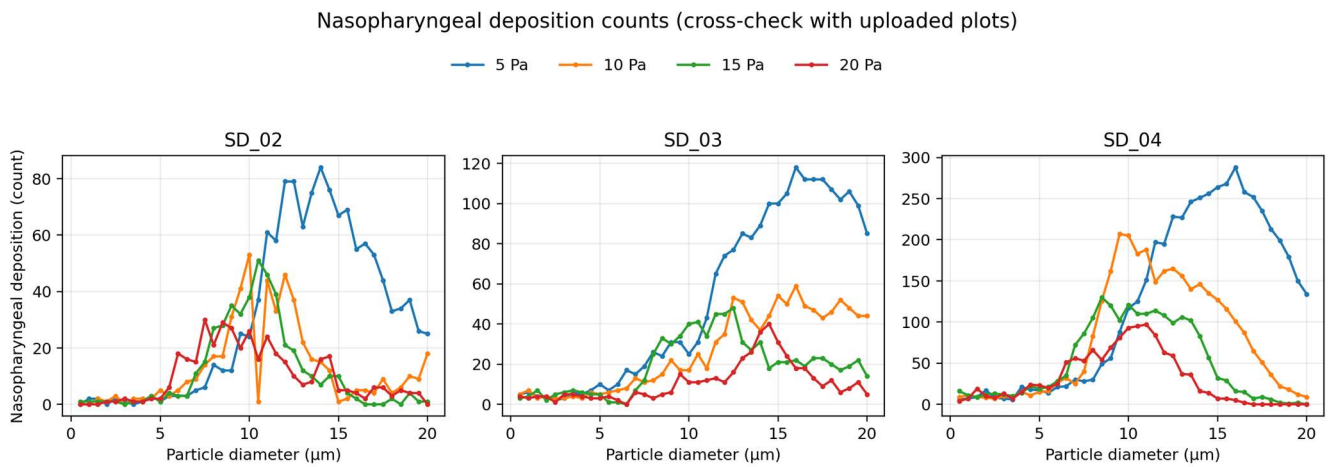

Figure 4. Workbook-derived nasopharyngeal counts, included as a consistency check against the uploaded nasopharyngeal deposition PDFs.

5. Engineering interpretation and limitations

From an engineering standpoint, the data indicate that anatomy controls not only how much deposition occurs, but where in the airway hierarchy the transition happens. SD_02 behaves like a geometry with relatively direct switching between penetration and nasal capture. SD_04 behaves like a stronger intermediate filter, especially for mid-size particles. SD_03 lies between those limits and may be the most pressure-sensitive geometry in the present set.

The main limitation is that the report is descriptive rather than statistical. No replicate simulations or uncertainty bands were provided, so differences should be interpreted as deterministic comparisons within the supplied dataset. A second limitation is nomenclature consistency: the workbook's lower-airway column and the uploaded escaped-particle plots should be reconciled in a future data curation step so that deposition, penetration, and escape are separated unambiguously.

6. Conclusions

- All three geometries exhibit a robust diameter-driven redistribution from downstream penetration at small sizes to dominant nasal capture at large sizes.
- Increasing pressure shifts this redistribution to smaller particles. The transition diameter falls from about 11.5–13.0 μm at 5 Pa to about 7.5–8.0 μm at 20 Pa.
- SD_04 is the strongest nasopharyngeal-filtering geometry, SD_02 is the weakest, and SD_03 is the intermediate case with the sharpest mid-range pressure sensitivity.
- For publication-quality reporting, normalized regional fractions should be used for comparison, while raw count plots should be retained as supporting trend evidence only.

Data source note: This report was generated from the uploaded workbook and the accompanying nasopharyngeal/escaped trend PDFs provided in the conversation.